
VAPO

Value-Model-Based Reinforcement Learning
for Advanced Reasoning

60.4 on AIME 2024

(avg@32) Surpassing BAPD (50) by 10+ points

ByteDance Seed · Base model: Qwen2.5-32B · No SFT data

arxiv.org/abs/2504.05118

Why Value Models?

Finer Credit Assignment for Long Chain-of-Thought Reasoning

Value-Model-Free (GRPO, DAPO)

- ▶ Compute advantage from final trajectory reward
- ▶ Assign advantage uniformly across all tokens
- ▶ Simple but coarse credit assignment
- ▶ Effective for shorter reasoning chains
- ▶ Limited for complex multi-step reasoning

Value-Model-Based (VAPO)

- ▶ **Credit assignment**
Token-level credit, tracing each action's impact on returns
- ▶ **Variance reduction**
Lower-variance per-token estimates vs Monte Carlo methods
- ▶ **Generalization**
Well-trained value model generalizes across samples

Critical when subtle step-level errors in long CoT cause catastrophic failures.

Three Core Challenges

Plaguing Value-Model-Based RL in Long Chain-of-Thought Tasks

C1

Value Model Bias

Long trajectories + bootstrapped learning → compounding errors in value estimates

Severity: High — renders value models unreliable without mitigation

C2

Heterogeneous Lengths

Short and long responses need different bias-variance tradeoffs for GAE λ parameter

Unique to long-CoT reasoning; ignored by standard methods

C3

Reward Signal Sparsity

Only the final <eos> token receives reward; all intermediate tokens get no direct signal

Exacerbated by long CoT — thousands of tokens, one reward

VAPO Framework: Integrated Solutions

Mapping Each Technique to the Challenge It Addresses

C1: Value Model Bias



Value-Pretraining

Pre-train value model before RL to reduce initial bias

Decoupled-GAE

Separate GAE computation for robust value estimation

C2: Heterogeneous Lengths



Length-Adaptive GAE

Dynamically adjust λ based on response length

Token-Level Loss

Normalize loss at token level, not sequence level

C3: Reward Sparsity



Clip-Higher

Asymmetric clipping to encourage exploration

Positive Example LM Loss

Self-imitation learning on correct trajectories

Group-Sampling

Multiple responses per prompt for better baselines

Length-Adaptive GAE

Dynamic λ Adjustment for Variable Response Lengths · Novel contribution (+15 pts)

Core Idea: Standard GAE uses a fixed λ across all sequences. But short and long responses have fundamentally different bias-variance characteristics. Length-Adaptive GAE dynamically tunes λ per response, optimizing the tradeoff for each trajectory.

Short Responses

$\lambda \rightarrow$ Lower

- ▶ Fewer tokens $\rightarrow$ less bootstrapping error
- ▶ Lower λ reduces variance
- ▶ Value estimates are more reliable here
- ▶ **Prioritize: LOW VARIANCE**

Long Responses

$\lambda \rightarrow$ Higher

- ▶ Many tokens $\rightarrow$ compounding bootstrap error
- ▶ Higher λ reduces bias from value model
- ▶ Trades some variance for accuracy
- ▶ **Prioritize: LOW BIAS**

Ablation: Removing Length-Adaptive GAE drops AIME 2024 from 60 $\rightarrow$ 45 (−15 points)

AIME 2024 Results

VAPO scores 60 — 10+ points above DAPO and DeepSeek-R1-Zero (avg@32, Qwen2.5-32B base)

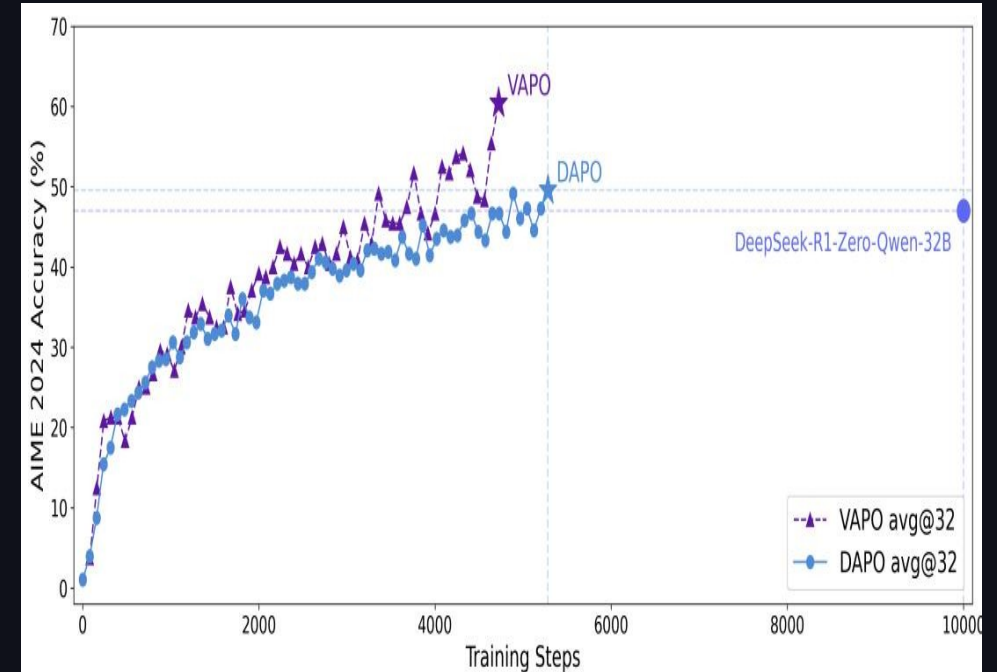
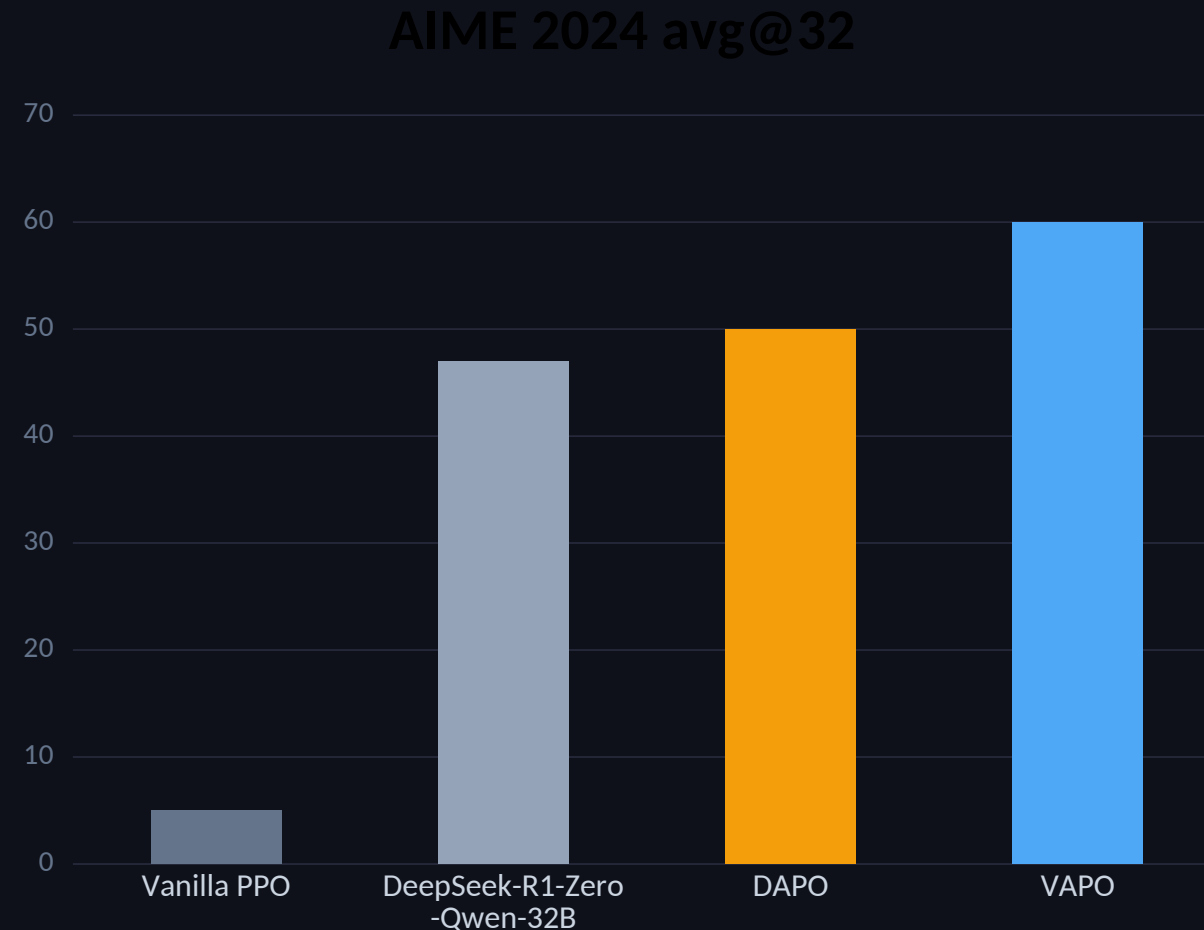


Figure 2: Training curve — VAPO vs DAPO over training steps

- ▶ VAPO reaches 60% at ~step 4,500
- ▶ DAPO plateaus near 50% at step 5,500
- ▶ DeepSeek-R1-Zero: 47% after ~10,000 steps

Training Efficiency & Stability

VAPO reaches SOTA in ~5,000 steps — zero training crashes across multiple runs

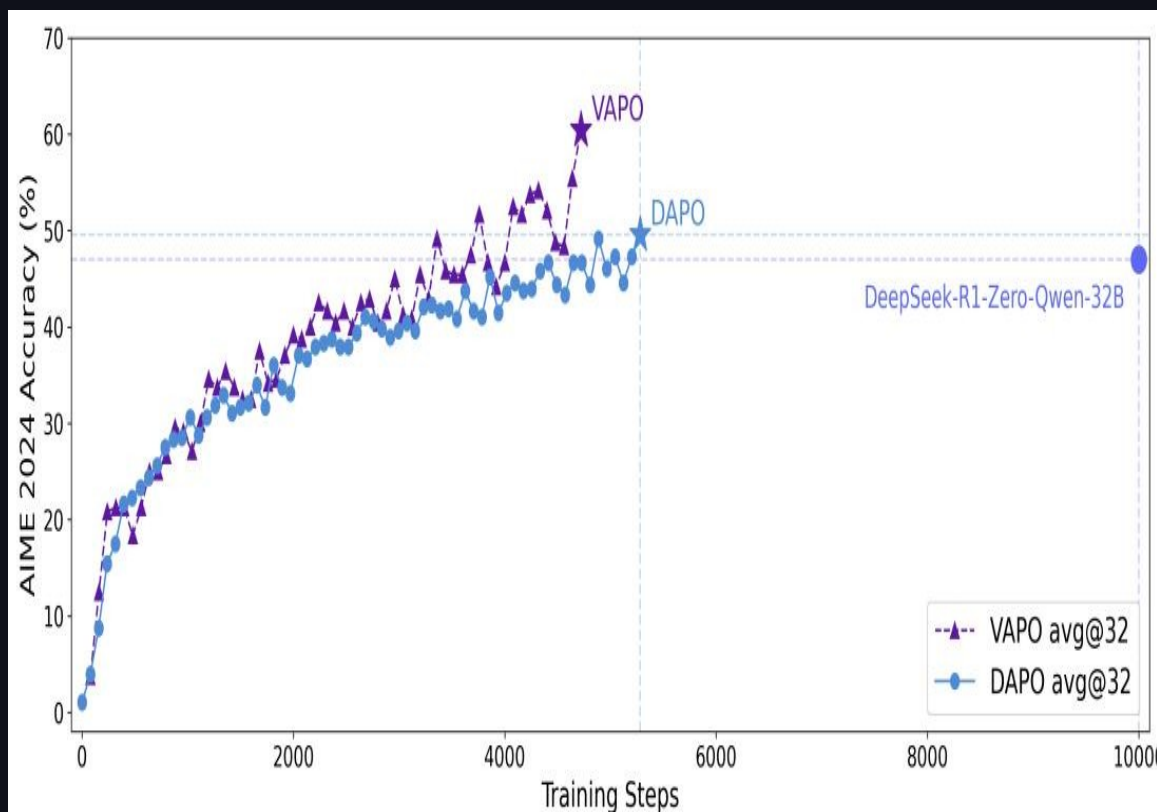


Figure 2: AIME 2024 accuracy over training steps

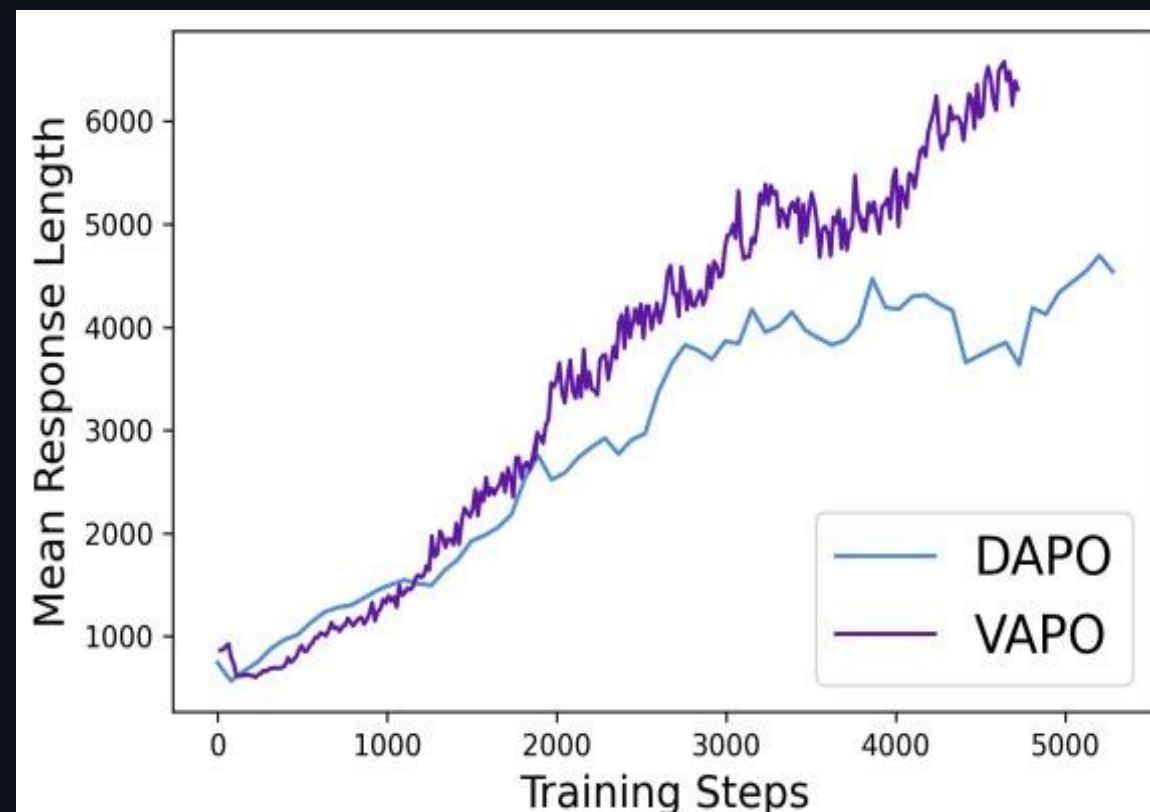
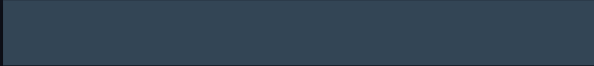


Figure 3: Mean response length over training

VAPO: ~5K steps to SOTA · DAPO: ~5.5K steps for 50% · DeepSeek-R1-Zero: ~10K steps for 47% · VAPO generates ~6,000 token responses vs DAPO's ~4,500

Ablation: Every Component Matters

Value-Pretraining alone accounts for 49 points · AIME 2024 avg@32

Configuration	Score	Δ from Full	Impact
VAPO (Full)	60	—	
w/o Group-Sampling	55	-5	
w/o Positive Ex. LM Loss	54	-6	
w/o Token-Level Loss	53	-7	
w/o Clip-Higher	46	-14	
w/o Length-Adaptive GAE	45	-15	
w/o Decoupled-GAE	33	-27	

Critical finding: Removing Value-Pretraining collapses performance from 60 → 11 (-49 pts). Value-model-based RL barely works without proper initialization.

Key Takeaways

Value-Model-Based RL Is Back — With the Right Engineering

Value-model-based RL surpasses value-free methods

VAPO achieves 60.4 on AIME 2024, surpassing DAPO (50) and DeepSeek-R1-Zero (47) by 10+ points — all on the same Qwen2.5-32B base with no SFT data.

Value-Pretraining is the single most critical component

Contributing 49 points (11 → 60), it is the prerequisite that makes value-model-based RL viable. Without it, performance collapses.

Length-Adaptive GAE solves the heterogeneous length problem

A novel technique contributing 15 points. Dynamic λ adjustment per response length optimizes bias-variance tradeoffs unique to long-CoT reasoning.

VAPO is training-efficient and stable

Reaches SOTA in ~5,000 steps with zero training crashes across multiple independent runs, compared to DAPO's ~5,500 steps for a lower score.

Implications for future reasoning models

Systematically addressing bias, heterogeneous lengths, and reward sparsity unlocks value-model-based RL — a design pattern likely to generalize beyond math reasoning.